

Student Performance Data Explorer

Exploratory Data Analysis Report | AI/ML Internship – Task 1

1. Problem Statement

Educational institutions collect student performance data but often lack meaningful insights into trends and learning outcomes. This analysis explores a student performance dataset to identify patterns in scores, study habits, and attendance, and presents the findings through visualizations.

2. Dataset Overview

The dataset contains 206 student records with the following fields: Student ID, Name, Gender, Study Hours, Attendance (%), Math Score, Science Score, and English Score. The raw data included missing values in five numeric columns, six duplicate rows, and inconsistent Gender labels (e.g. 'male', 'Male', 'female '). These issues were resolved during the cleaning stage: missing values were filled using each column's median, duplicates were removed, and text labels were standardized.

3. Statistical Summary

After cleaning, an **average_score** column was added as the mean of Math, Science and English scores per student. Math showed the highest average score among the three subjects, while English showed the widest spread in values.

4. Score Distributions

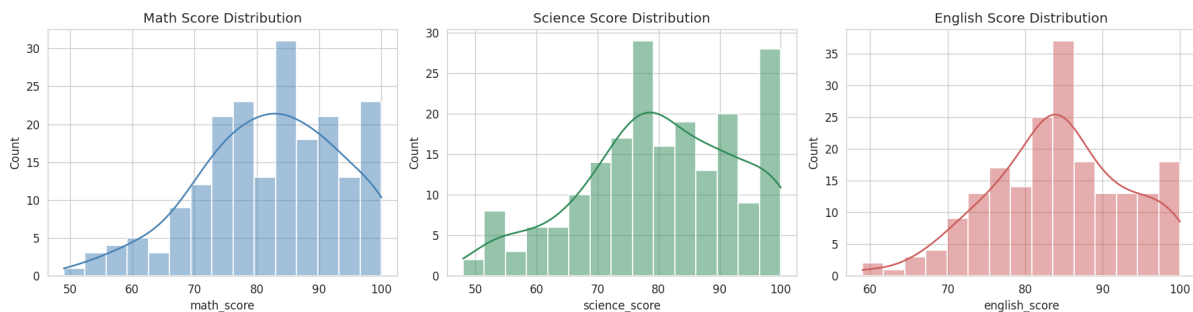


Figure 1: Distribution of Math, Science and English scores.

5. Outlier Analysis (Box Plots)

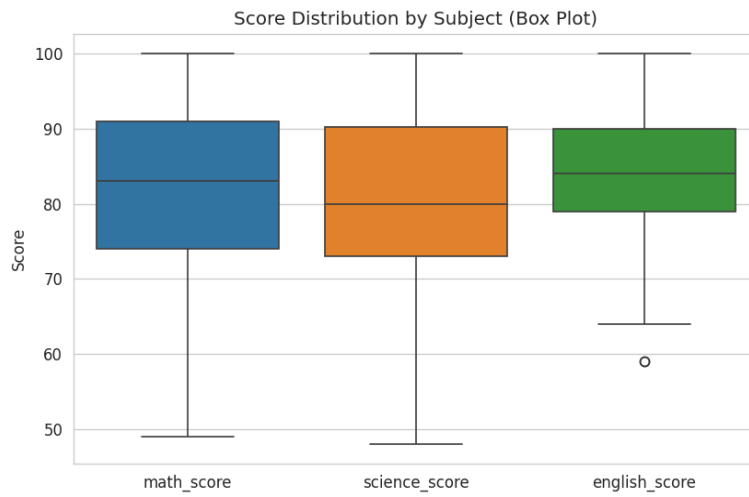


Figure 2: Score spread and outliers by subject.

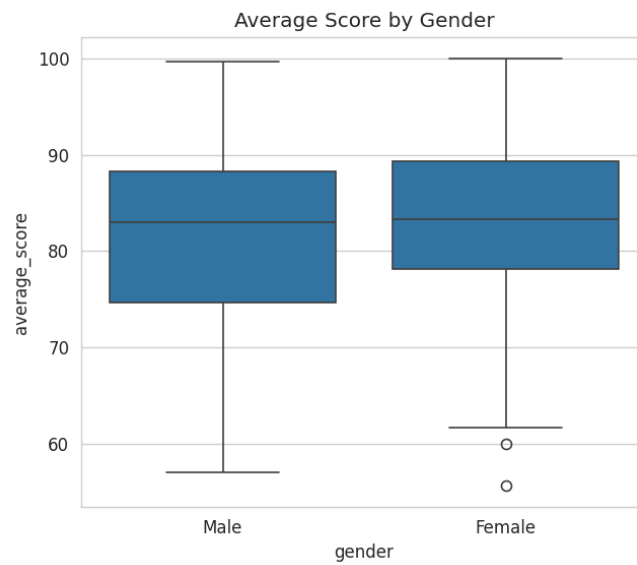


Figure 3: Average score by gender.

6. Relationships: Study Habits vs Performance

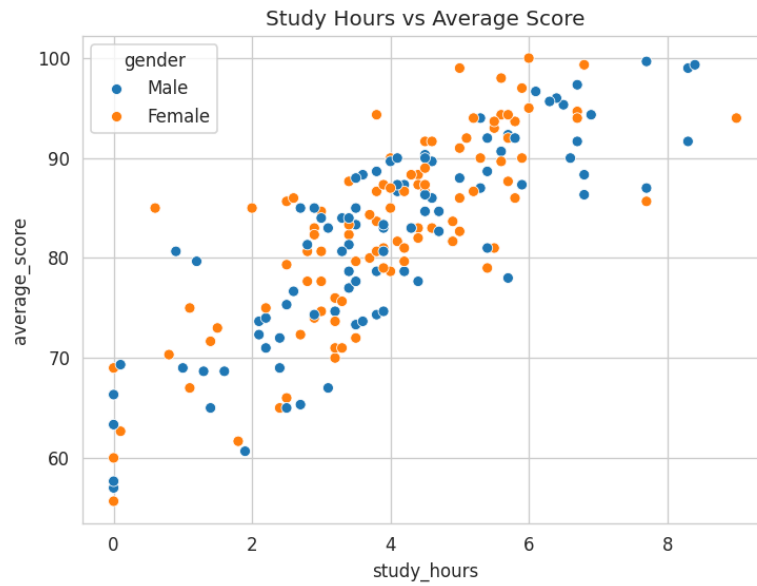


Figure 4: Study hours vs average score.

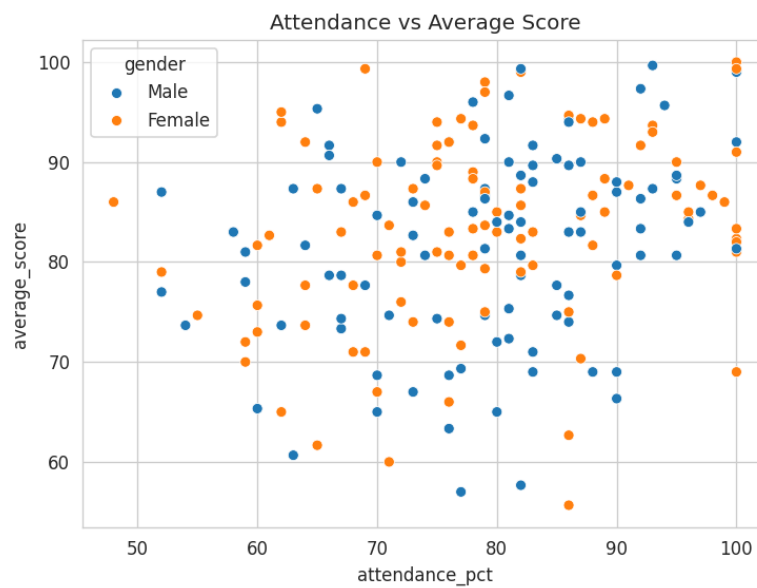


Figure 5: Attendance vs average score.

7. Correlation Heatmap

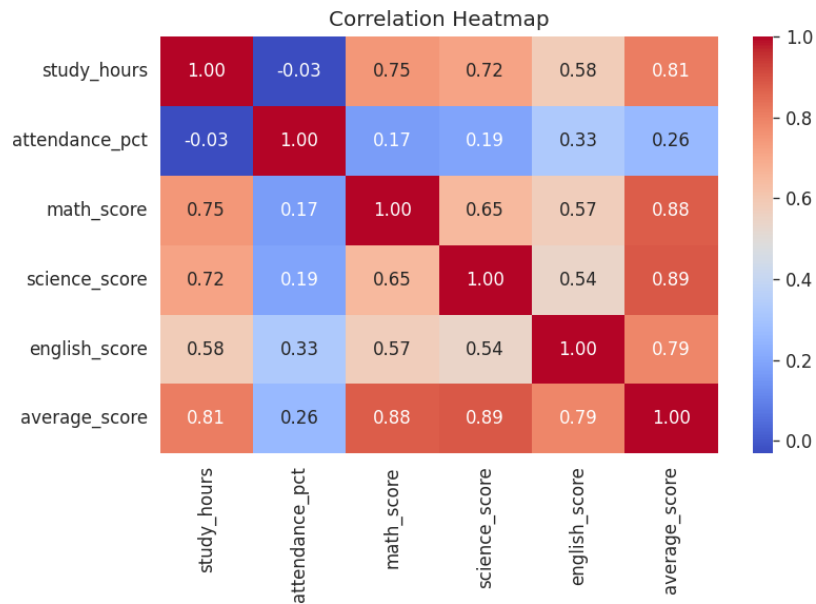


Figure 6: Correlation between study hours, attendance and scores.

8. Key Findings

- Study hours and attendance both show a positive relationship with average score.
- Math has the highest average score among the three subjects; English shows the widest spread.
- Attendance correlates with scores slightly more strongly than study hours do, suggesting consistent class attendance matters as much as self-study.
- Gender does not show a large difference in average score between groups.
- A small number of outliers exist in each score column; these were reviewed individually rather than removed automatically.

9. Recommendation

Attendance and study hours together are reasonable early indicators of performance and could be monitored to flag at-risk students. This cleaned and analyzed dataset is ready as a baseline for predictive modeling in Task 2.